MILITARY SPECIFICATION

TRAINING SET, CHEMICAL AGENT IDENTIFICATION, SIMULANTS, M72A2

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a simulants chemical agent identification training set which contains no chemical agents.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-676 - Boxes, Setup.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-51409 - Paper, Chemical Agent Detector, VGH, M8, Booklet of.

FSC 6910

: Beneficial comments (recommendations, additions, deletions) and any :
: pertinent data which may be of use in improving this document should :
: be addressed to: Commander, US Army Armament Research and Development:
: Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by :
: using the self-addressed Standardization Document Improvement Proposal:
: (DD Form 1426) appearing at the end of this document or by letter. :

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D5-77-2350 - Training Set, Chemical Agent Identification, Simulants, M72A2.

PUBLICATIONS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY PURCHASE DESCRIPTION

EA-D-1155 - Detector Kit, Chemical Agent: M256.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials cited on Drawing D5-77-2350 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Components. All components of the training set shall conform to the specifications and drawings listed on Drawing D5-77-2350 and subsidiary drawings.

3.3 Manufacture and assembly. The training set shall be manufactured and assembled in accordance with Drawing D5-77-2350 and subsidiary drawings.

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:00Z
Check the source to verify that this is the current version before use.

3.4 Functioning. The simulant from each simulant bottle and from each ampoule shall produce the required color change when tested as specified in 4.4.4.1.

3.5 Ampoule leakage. Each type of ampoule shall show no evidence of leakage when tested as specified in 4.4.4.2.

3.6 Workmanship. The training set shall be free from foreign matter such as grease, oil, and dirt. The training set shall be free from visible damage such as cracks and dents.

3.7 Preproduction. Prior to the start of regular production, a preproduction sample of training sets shall be produced in accordance with this specification for examination and test (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.1 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five training sets shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The sample training sets shall be forwarded for tests to the test facility specified in the contract (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the current version before use.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The five sample training sets shall be examined for the defects listed in 4.4.3.1.

4.3.2.2 For test. The five sample training sets shall be tested for functioning as specified in 4.4.4.1. Five of each type of ampoule shall be tested per set. All the ampoules in the training sets shall be tested for leakage as specified in 4.4.4.2.

4.3.3 Acceptance/rejection criteria. The presence of any examination defect in any of the preproduction samples or the failure of the functioning test or leakage test by any of the preproduction samples shall be cause for rejection of the preproduction lot.

4.4 Quality conformance inspection.

4.4.1 Lotting.

4.4.1.1 Components. A lot shall consist of like components manufactured by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.1.2 Training sets. A lot shall consist of the training sets manufactured by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of each type of simulant shall be represented in any one lot of finished training sets.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test.

(a) Functioning. Sampling shall be conducted in accordance with table I.

(b) Ampoule leakage. Sampling for each type of ampoule shall be conducted separately in accordance with MIL-STD-105 using an AQL of 0.65 percent defective.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:00Z
Check the source to verify that this is the currentversion before use.

TABLE I. Sampling for functioning test

Number of training sets in lot	Number of sample training sets to be tested
2 to 15	2
16 to 25	3
26 to 90	5
91 to 150	8
151 to 500	13
501 to 1,200	20
1,201 to 10,000	32
10,001 to 35,000	50
over 35,000	80

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample training sets and the preparation for delivery thereof shall be examined in accordance with MIL-STD-105, inspection level S-4 and the following classification of defects:

(a) Training set, chemical agent identification, simulants, M72A2 (Drawing D5-77-2350).

Categories	Defects
Critical:	None defined
Major:	AQL 0.65 percent defective
101	Component missing
102	Component incorrectly assembled
103	Workmanship (see 3.6)

(b) Test bottle assembly (Drawing C5-77-2362).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Bottle dimensions not as specified
102	Marking on bottle or cap incorrect, missing, or illegible

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:00Z
Check the source to verify that this is the current version before use.

(c) Simulant bottle assembly (Drawing C5-77-2148).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Bottle dimensions not as specified
102	Contents missing
103	Cap not securely tightened
104	Marking on bottle or cap incorrect, missing, or illegible

(d) Vial carrier (Drawing C5-77-2355).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Dimensions incorrect

(e) Simulant and test bottle carrier (Drawing C5-77-2357).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Dimensions incorrect

(f) Sampling adapter carrier (Drawing C5-77-2364).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Dimensions incorrect

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the current version before use.

(g) Ampoule assembly (Drawing C5-77-2359).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Length incorrect
102	Fill volume incorrect

(h) Adapter, sampling assembly (Drawing C5-77-2353).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Component missing
102	Assembled incorrectly
103	Dimensions incorrect
104	Workmanship (see 3.6)

(i) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Contents of vial boxes, carriers, or pail missing or not as specified
102	Vials of ampoules in paperboard box not packaged as specified
103	Arrangement of empty test bottles and filled simulant bottles in carrier not as specified
104	Packing not as specified
105	Pail or pail closure not as specified
106	Cushioning material missing or not as specified (where needed)
107	Fiberboard packing aids missing or not as specified
108	Marking of vial boxes incorrect, missing or illegible
109	Marking of pail incorrect, missing, or illegible

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the current version before use.

4.4.3.2 For test.

(a) Functioning. The components taken as specified in 4.4.2.2(a) shall be tested as specified in 4.4.4.1. One of each type of ampoule shall be tested per set. Failure of any component to produce the required color change shall be cause for rejection of the lot represented.

(b) Ampoule leakage. The sample ampoules taken as specified in 4.4.2.2(b) shall be tested as specified in 4.4.4.2.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning.

(a) Test bottles and ampoules. Using the necessary components from an M256 detector kit conforming to Purchase Description EA-D-1155, sample and test the vapor from each type of ampoule using the applicable test bottle in the training set and observe for the required color change in accordance with the applicable instructions provided with the M256 detector kit and the instructions in the technical manual for the training set.

(b) Simulant bottles. Test each type of simulant bottle in the training set as follows: Using the applicator in the vial cap, place one drop of simulant on M8 detector paper conforming to MIL-P-51409 and observe for the required color change in accordance with the instructions provided with the M8 detector paper and the instructions in the technical manual for the training set. Replace the cap.

4.4.4.2 Ampoule leakage. Place the ampoule in a vacuum desiccator and reduce the pressure to 127 millimeters (mm) of mercury absolute or less. Close the stopcock of the desiccator, maintain the vacuum for 10 minutes, and then draw in sufficient colored water to cover the ampoule. Discontinue the vacuum and allow the ampoule to remain in the colored water for 5 minutes. Examine the ampoule for evidence of leakage. Penetration of colored water into the ampoule shall be considered evidence of leakage.

5. PACKAGING

5.1 Packaging, level A. Each of the five types of ampoules shall be packaged as follows: Fifty vials containing the same type of ampoules shall be packaged with the closure end upright in a setup paperboard box conforming to type I, variety 1, class C, style 4 of PPP-B-676. The inside dimensions of the box shall be 101.6 mm by 50.8 mm by 50.8 mm for length, width, height, respectively. Fiberboard pads shall be inserted

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

flat against the inner faces of the ends of the box in sufficient quantity to prevent movement of the vials. The pads shall be formed from material conforming to type CF, class domestic, variety and grade optional of PPP-F-320 and shall be 63.5 mm long by 50.8 mm wide. The pads shall be placed with the longer dimension upright in the box.

5.2 Packing, level A. Three filled molded polystyrene carriers shall be placed in a pail as shown on Drawing D5-77-2350. The bottom carrier (Drawing D5-77-2354) shall contain a sampling adapter assembly (Drawing C5-77-2353). The middle carrier (Drawing C5-77-2356) shall contain five boxes of vials packaged as specified in 5.1. All boxes of vials shall be packed so that the marked end of the box is visible. The top carrier (Drawing C5-77-2360) shall contain three filled simulant bottles (Drawing C5-77-2148) and eight empty test bottles (Drawing C5-77-2362). Cushioning material shall be placed in the carriers as needed to prevent movement of contents and to insure a tight fit. A sufficient quantity of fiberboard conforming to type CF, class domestic, grade 125 of PPP-F-320 shall be placed on top of the top carrier to insure a tight fit. A copy of the technical manual for the training set shall be placed on top of the fiberboard and the other contents of the pail.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. The top of the cover and one end of each box of vials packaged as specified in 5.1 shall be marked for identification with black printer's ink by showing the symbol (minimum 6.35 mm letters) corresponding to the identification marking on the contents. The pail shall be marked as shown on Drawing D5-77-2363.

5.3.1 Precautionary marking. The top cover of each box of vials shall be marked as follows:

WARNING

Harmful if inhaled or swallowed.
Avoid contact with eyes, skin, and clothing.
Use with adequate ventilation.
Wash thoroughly after handling.

6. NOTES

6.1 Intended use. The training set covered by this specification is intended for use in training personnel in the identification of certain chemical agents.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6910-A373

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1426 JAN 66 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DOD-314

OFFICIAL BUSINESS

Commander
US Army Armament Research
and Development Command
Attn: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.